

BONUS CONTENT

"FEEDBACK"

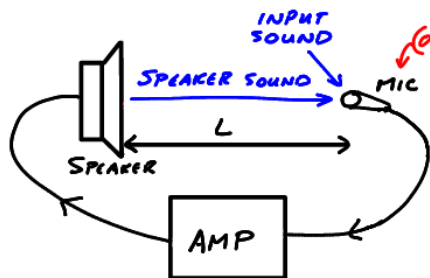
- BASICS
- RINGING  DEMO HENDRIX "FOXEY LADY" INTRO RINGING
- BRASS INSTRUMENTS
- REED INSTRUMENTS
- EDGE INSTRUMENTS
- CONES + HARMONICS
- VOICE?

FEEDBACK BASICS

FEEDBACK IS EVERYWHERE, FROM HELPING YOU & ROBOTS WALK TO SOUND GENERATION TO THE ANNOYING SCREEEE ON STAGE.

WHAT PEOPLE USUALLY THINK OF...

-CONSIDER A MICROPHONE, AMPLIFIER, & SPEAKER IN A ROOM:

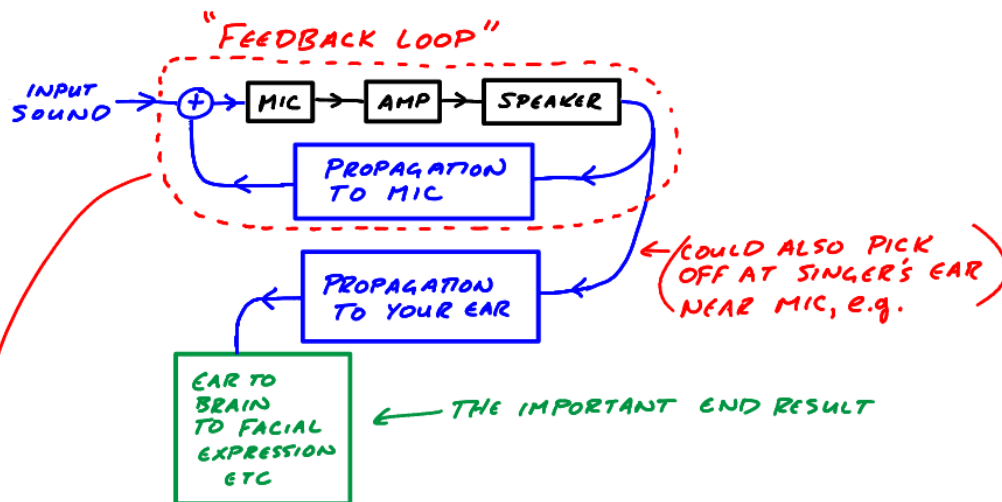


1. MICROPHONE CONVERTS SPEAKER + INPUT SOUND TO VOLTAGE $\rightarrow$ AMP
2. AMP BOOSTS VOLTAGE $\rightarrow$ SPEAKER
3. SPEAKER EMITS SOUND $\rightarrow$ MICROPHONE.

-THE QUESTION IS HOW WILL THIS WHOLE SYSTEM RESPOND TO INPUT SOUND?

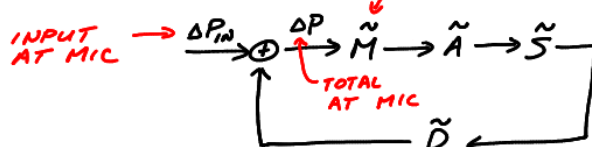
- WILL THIS LOOP AFFECT THE SOUND?

-TO ANALYZE THIS MATHEMATICALLY, WE LIKE TO DRAW "LOOP DIAGRAMS", e.g.



MATHEMATICALLY,

"~" MEANS "TRANSFORMATION" (JUST "SOMETHING HAPPENS TO SIGNAL")



ALL OF THESE ARE "SUSCEPTIBILITIES" WITH FREQUENCY-DEPENDENT GAIN AND PHASE, INCLUDING DELAYS.

-SYMBOLS

- $\rightarrow \tilde{M}$: MIC CONVERTS PRESSURE TO VOLTAGE $[V/P_a]$
- $\rightarrow \tilde{A}$: AMP CONVERTS VOLTAGE TO BIGGER VOLTAGE $[V/V]$
- $\rightarrow \tilde{S}$: SPEAKER CONVERTS VOLTAGE TO PRESSURE $[P_a/V]$
- $\rightarrow \tilde{D}$: PROPAGATION CONVERTS PRESSURE FROM SPEAKER TO PRESSURE AT MIC $[P_a/P_a]$
- $\oplus$: SUM OF ALL PRESSURES AT MIC



WHICH ELEMENT HAS THE GREATEST DELAY?

- IN GENERAL, THESE SUSCEPTIBILITIES TRANSFORM THE AMPLITUDE AND PHASE OF EACH FREQUENCY COMPONENT

EXAMPLE: $\Delta P = \Delta P_0 \cos(2\pi f t)$ AT MIC BECOMES

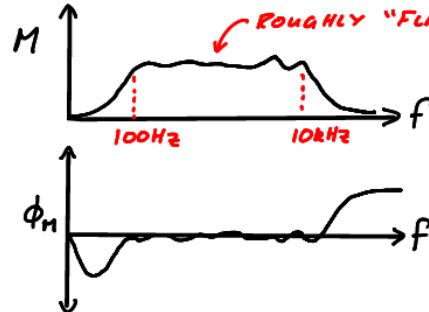
$$\rightarrow V = M(f) \Delta P_0 \cos(2\pi f t + \phi_m(f)) \quad \text{AT AMP INPUT}$$

$M(f)$ $\rightarrow$ f-DEPENDENT "GAIN" WITH UNITS $[V/P_a]$
 $\phi_m(f)$ $\rightarrow$ f-DEPENDENT PHASE SHIFT
 THESE FULLY DEFINE $\tilde{M}$

$M(f) + \phi_m(f)$ CAN DEPEND ON

- GEOMETRY, MASS, + STIFFNESS OF THE MICROPHONE'S MECHANICAL SENSOR
- FILTER MATERIALS SURROUNDING THIS SENSOR
- MECHANICAL RESONANCES OF THE WHOLE STRUCTURE (HANDLE, SCREEN, "...")
- ELECTRO-MECHANICAL CONFIGURATION + CIRCUIT.

TYPICAL SUSCEPTIBILITY FOR A VOCAL MIC



WE WON'T DELVE TOO DEEPLY INTO THIS BUT THERE ARE PHYSICAL RESTRICTIONS ON THE RELATIONSHIP BETWEEN $M(f) + \phi_m(f)$

- AMPS CAN BE MUCH MORE "FLAT" (ELECTRONS ARE FASTER THAN ATOMS)
- SPEAKERS USUALLY MUCH LESS FLAT
- WITHOUT DIGITAL BUFFERS, DELAYS IN ELECTRONICS ARE SHORT

20pt

Q

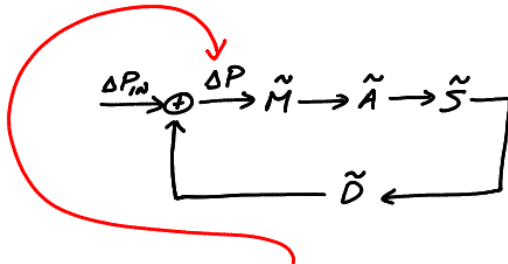
WHAT WILL INFLUENCE $D(f) + \phi_D$

DISTANCE TO MIC
 SHAPE OF ROOM
 WALL MATERIALS
 LOCATION OF SINGER
 SOUND WAVES FROM SPEAKER
 AMPLIFIER GAIN

30pt

Q

IF THE INPUT TO $\tilde{M}$ IS $\Delta P_m \cos(2\pi f t)$
WHAT IS THE OUTPUT OF $\tilde{A}$?



- SO WHAT IS ΔP IF $\Delta P_{IN} = \Delta P_i \cos(2\pi f t)$? (i.e., THE BIG QUESTION!)

- FOLLOWING THE LOOP,

$$\Delta P = \tilde{D}\tilde{S}\tilde{A}\tilde{M}\Delta P + \Delta P_{IN}$$

SIGNAL WE INJECT

THESE ARE THE ABOVE OPERATIONS (GAINS + PHASES) IN ORDER AS THE SIGNAL MAKES A COMPLETE LOOP.

ASSUME A PURE TONE RESPONSE OF UNKNOWN AMPLITUDE + PHASE

$$\Rightarrow \Delta P_o \cos(2\pi f t + \phi_o) = \tilde{D}\tilde{S}\tilde{A}\tilde{M}\Delta P_o \cos(2\pi f t + \phi_o + \phi_D + \phi_S + \phi_A + \phi_M) + \Delta P_i \cos(2\pi f t)$$

VARIABLES WE WISH TO CALCULATE

- APPROACH: ISOLATE $\cos(2\pi f t)$ + $\sin(2\pi f t)$ WITH TRIG + SOLVE FOR $\Delta P_o + \phi_o$

$\Rightarrow$ TEDIOUS, BUT NOT IMPOSSIBLE (BONUS?)

- ENTHUSIASTS: WITH $\Delta P = \Delta P_o e^{i2\pi f t + \phi_o}$ AND $\Delta P_{IN} = \Delta P_i e^{i2\pi f t}$

$\Rightarrow \tilde{M}, \tilde{A}, \tilde{S}, \tilde{D}$ ARE JUST COMPLEX FACTORS THAT MULTIPLY!!! ($Ae^{i\phi} Be^{i\psi} = ABe^{i(\phi+\psi)}$)

$$\Rightarrow \Delta P = \tilde{D}\tilde{S}\tilde{A}\tilde{M}\Delta P + \Delta P_{IN}$$

$$\Rightarrow \Delta P = \left(\frac{1}{1 - \tilde{D}\tilde{S}\tilde{A}\tilde{M}} \right) \Delta P_{IN}$$

COMPLEX "CLOSED LOOP GAIN" FACTOR, COMPLETE WITH MAGNITUDE + PHASE!!

$\Rightarrow$ DONE

OR "GAIN"

- IN GENERAL THIS "CLOSED-LOOP" SUSCEPTIBILITY TO EXTERNAL INPUT IS WELL BEHAVED, UNTIL AN IMPORTANT CONDITION IS MET:

DOESN'T EXPLODE

$\phi_D + \phi_S + \phi_A + \phi_M = 0, 2\pi, 4\pi, \dots$ AT "SPECIAL" FREQUENCIES

- AT THESE "SPECIAL" FREQUENCIES

AMPLITUDE GAINS MULTIPLY

OTHER ADD PHASES TO $0, \pm 2\pi, \dots$

$$\Rightarrow \Delta P_o \cos(2\pi f t + \phi_o) = \tilde{D}\tilde{S}\tilde{A}\tilde{M}\Delta P_o \cos(2\pi f t + \phi_o) + \Delta P_i \cos(2\pi f t)$$

- FOR THIS TO BE TRUE AT ALL TIMES, $\phi_o = 0, \pm 2\pi, \pm 4\pi, \dots$ AND

$$\Delta P_o = \tilde{D}\tilde{S}\tilde{A}\tilde{M}\Delta P_o + \Delta P_i \quad (\text{A RELATIONSHIP BETWEEN AMPLITUDES ONLY})$$

$$\Rightarrow \Delta P_o = \underbrace{\left(\frac{1}{1 - DSAM} \right)}_{\text{20pt Q}} \Delta P_i$$

20pt
Q

AT THIS "SPECIAL" FREQUENCY
WHAT HAPPENS IF THE "OPEN-LOOP GAIN" Δ DSAM
IS CLOSE TO 1 (AND TOTAL LOOP PHASE = 2π)?

20pt
Q

WHAT IS THE GAIN AND PHASE ASSOCIATED WITH A DELAY Δt ?

$$\Delta P_o \cos(2\pi f t) \rightarrow$$

- DELAY TRANSFORMATION IS JUST A PHASE THAT INCREASES WITH f ↗

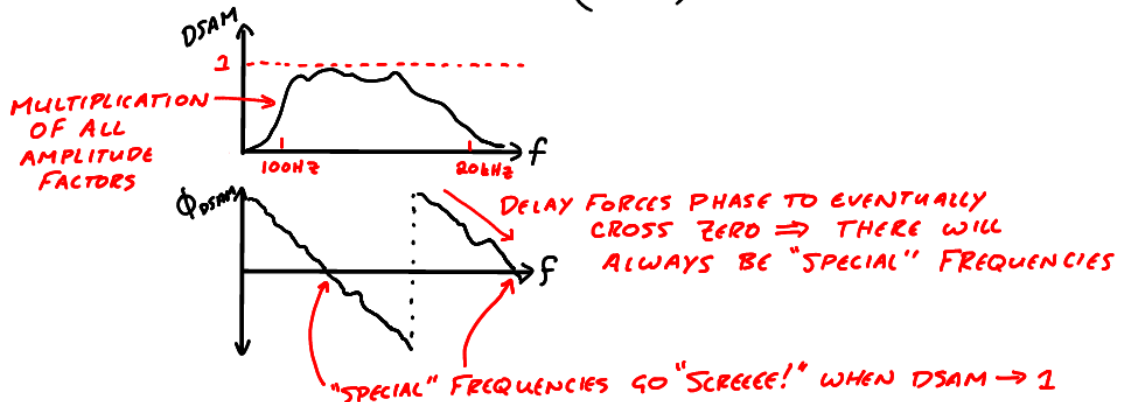
DEMO

DESMOS 7.1 FEEDBACK

- SHOW DELAY SHIFTING THE CURVE
- SHOW FREQUENCY AFFECTS RELATIVE PHASE
- FIND FREQUENCY WHERE RINGING IS EXPECTED

DELAY $\Rightarrow$ THE CLOSED-LOOP GAIN WILL EXPLODE AT SPECIFIC FREQUENCIES, NO MATTER WHAT, IF YOU INCREASE GAIN TOO FAR.

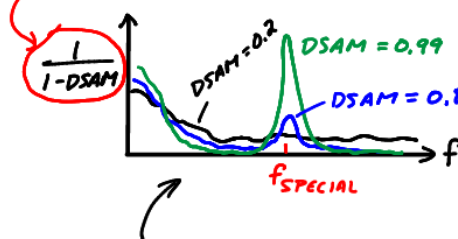
- TYPICAL "OPEN-LOOP" GAIN (DSAM)



- CHANGING DELAY (MOVING MIC) CHANGES THE SCREEEE PITCH
- ALSO AFFECTED BY RESONANCES IN THE ROOM ETC SO USUALLY NOT WELL-BEHAVED

BONUS: MAKE A MIC + SPEAKER FEEDBACK INSTRUMENT !!

- IN PRACTICE, AS WE TURN UP THE GAIN AND DSAM APPROACHES 1 AT f_{SCREEN} , WE SEE A GROWING PEAK IN THE "CLOSED-LOOP" GAIN AT f_{SCREEN} UNTIL IT FINALLY RUNS AWAY



THE WHOLE SYSTEM STARTS TO LOOK LIKE A RESONATOR!
 $\Rightarrow$ YOU WILL HEAR SPECIFIC FREQUENCIES RINGING FOR AWILE WHEN YOU TALK, ETC...

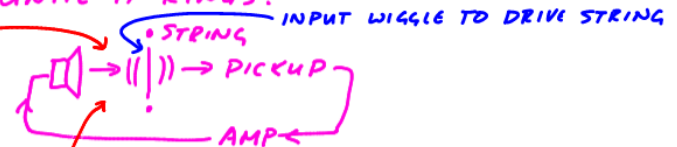
LIVE SOUND ENGINEERS WORK VERY HARD TO PREVENT THESE "CLOSED-LOOP RESONANCES"

THIS ALSO HAPPENS WHEN HENDRIX HOLDS HIS GUITAR TO THE AMP, BUT STRING RESONANCES + PERHAPS SOME NON-AUDIO/INDUCTIVE PLAY A ROLE!

DEMO BEGINNING OF "FOXY LADY"

- HE "SEEDS" THE STRING RESONANCE WHILE APPROACHING THE AMP TO INCREASE THE OPEN-LOOP GAIN ($\frac{1}{R^2}$ SPREAD) UNTIL IT RINGS.

SET THE DELAY WITH DISTANCE TO SELECT STRING RESONANCE



(THERE CAN ALSO BE DIRECT INDUCTIVE PICKUP VIA MAGNETIC FIELDS)

THE STAGE + YOU ON IT ARE PART OF THE INSTRUMENT ☺

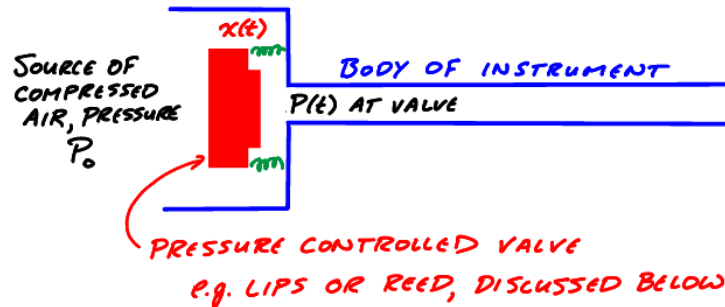
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Q IGNORING $f=0$ (WHERE GAIN $\rightarrow \infty$) WHAT SPEAKER-MIC DISTANCE "D" CAUSE RINGING AT 340 Hz?

FEEDBACK IN MUSICAL INSTRUMENTS

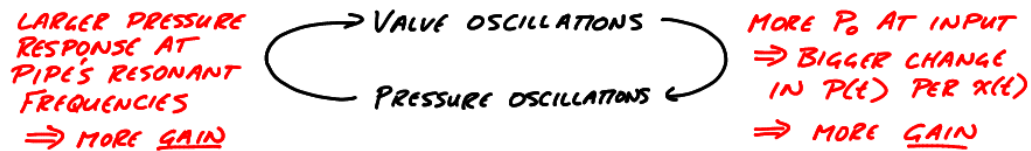
PRESSURE-CONTROLLED VALVE

- CONSIDER THIS TOY MODEL FOR MANY WINDINSTRUMENTS



CLOSES IF $P_0 > P(t)$
OPENS IF $P_0 < P(t)$, LETS IN AIR, RAISES $P(t)$

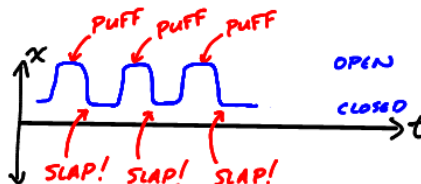
THIS IS ANOTHER FEEDBACK LOOP:



- THIS PROCESS NATURALLY SELECTS FREQUENCIES NEAR THE INSTRUMENT'S BODY RESONANCES.

- BY ADJUSTING THE SPRING STIFFNESS AND MASS (TIGHTENING LIPS, e.g.), AND CHANGING THE INPUT PRESSURE, WE TUNE THE VALVE'S SUSCEPTIBILITY AND CAN CHANGE WHICH FREQUENCY IS NATURALLY AMPLIFIED.

- USUALLY IS HIGHLY NONLINEAR WITH THE VALVE SLAPPING SHUT EVERY CYCLE ⇒ COMPLICATED

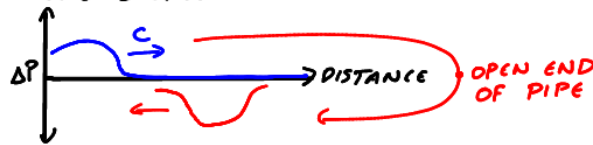


⇒ IT'S HELPFUL TO THINK OF PULSES!

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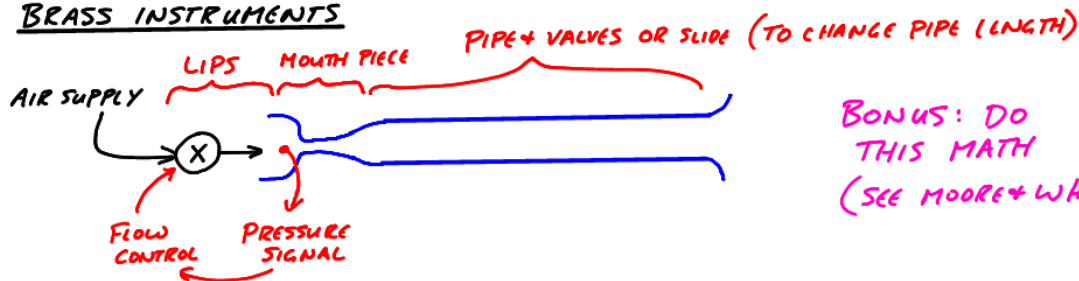
Q

IF THE VALVE OPENS, A HIGH PRESSURE FRONT RUNS DOWN THE BODY...



WHAT IS THE REFLECTED PRESSURE FRONT?

BRASS INSTRUMENTS



BONUS: DO THIS MATH (SEE MOORE + WHEELER?)

- PRESSURE SIGNAL FROM PIPE RESONANCES CAN REINFORCE LIP OSCILLATIONS

⇒ CAN TIGHTEN LIPS TO MAKE ANY TONE ON YOUR OWN, BUT INSTRUMENT RESONANCES + FEED BACK

MAKES TRANSFER TO SOUND WAVE ENERGY MORE EFFICIENT { TEND TO "LOCK" TO SPECIFIC FREQUENCIES, ENHANCE THE PRESSURE WAVE AMPLITUDE, AND FILTER OUT THE SOUND (KILL THE HIGH-FREQUENCY "BUZZING" HARMONICS)

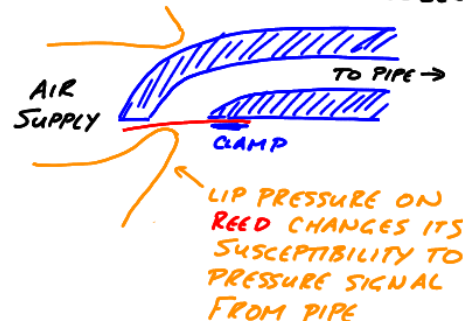
- LIPS ARE A LOT LIKE VOCAL CORDS, JUST NEEDING AN EXTERNAL RESONATOR (NO MOUTH, SINUSES, ETC)

REEDS: THIN, STIFF PIECE OF MATERIAL

⇒ LIGHT-WEIGHT HARMONIC OSCILLATOR WITH HIGH RESONANT FREQUENCY (MUCH HIGHER THAN THE PLAYED NOTES)

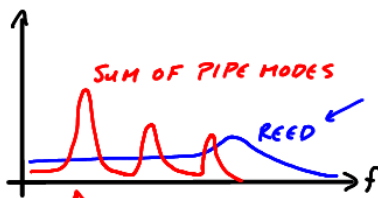
- FLICK THEM + LISTEN, OR PLAY THEM WITHOUT THE INSTRUMENT!

- REEDS ALSO ACT AS A PRESSURE-CONTROLLED VALVE



- AGAIN, RESONANCES IN THE PIPE FEED BACK TO THE REED, ENHANCING SPECIFIC FREQUENCIES

- THESE CAN BE MUCH LOWER THAN THE REED'S NATURAL FREQUENCY, WHICH JUST MEANS THE REED OPERATES IN THE "FLAT" PART OF ITS OWN SUSCEPTIBILITY



USE LIPS TO SET A NATURAL "SLAP FREQUENCY" FOR A GIVEN AMPLITUDE (SLAPITUDE?) + DRIVE ENOUGH AIR TO ACHIEVE THAT AMPLITUDE STEADILY

DOWN HERE, THE REED JUST ACTS AS A SPRING CONSTANT
 $\Rightarrow$ PRESSURE & DISPLACEMENT IN PHASE WITH PRESSURE SIGNAL

BONUS: ANALYZE THIS

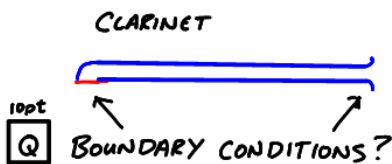
DEMO

CUT STRAWS

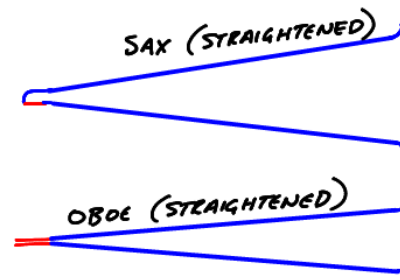
- NATURAL REED f_0 ?
- WITH PIPE?

- REED INSTRUMENTS TEND TO HAVE MORE HARMONICS (MORE "BITE" TO THE TIMBRE, BECAUSE THEY ARE LIGHT WEIGHT, HAVE A HIGHER NATURAL FREQUENCY, AND "CLICK" MORE WHEN THEY SLAP SHUT.

- BUT WHY DOES A CLARINET SOUND SO DIFFERENT FROM A SAX OR OBOE?

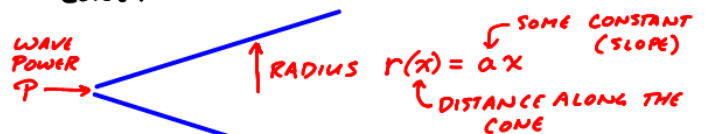


10pt
 Q RESONANT FREQUENCIES?



- THIS LOOKS LIKE THE SAME SITUATION AT FIRST, BUT WAVES IN A CONE BEHAVE DIFFERENTLY!

- CONSIDER A WAVE IN THIS "ALMOST CLOSED" CONE:



20pt
 Q FOR A WAVE CARRYING POWER P DOWN THE CONE, WHAT IS THE INTENSITY AT x ?

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 Q HOW DOES THE PRESSURE AMPLITUDE FALL OFF WITH x ?

LEFT OR RIGHT MOVING!

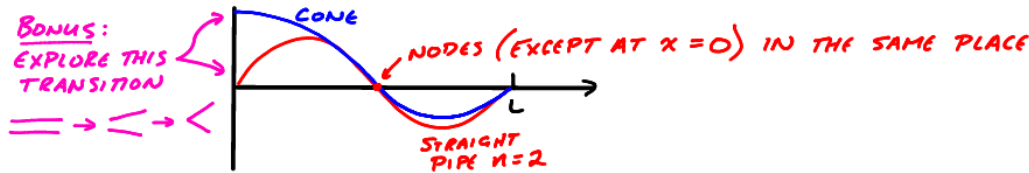
- THIS x -DEPENDENCE CHANGES THE WAVEFORM OF PURE TONES!

⇒ STANDING WAVES (PRESSURE NODE AT THE WIDE AND TINY ENDS)
HAVE A SPATIAL PROFILE

$$\propto \frac{1}{x} \sin(2\pi x_n x)$$

BUT THIS DOES NOT TEND
TOWARD A NODE AS YOU BRING
THE CONE TO A SMALLER & SMALLER
POINT!

DESMOS 7.2 CONE VS PIPE FOR $n=1, 2, 3, 4, \dots$



SO A PERFECT CONE SUPPORTS ALL HARMONICS
JUST LIKE A SYMMETRIC STRAIGHT PIPE!

- VERY COOL THAT THE NODE OF AN OPEN PIPE
TRANSFORMS INTO AN ANTINODE WHEN WE
MORPH IT INTO A CLOSED CONE (WHERE PRESSURE ISN'T VENTED)

- THIS $\frac{1}{x}$ MODIFICATION OBVIOUSLY MUST HAPPEN TO SOME EXTENT IN
THE MOUTHPIECE OF A CLARINET, BUT THE VAST MAJORITY OF THE PIPE
IS STRAIGHT, SO IT'S JUST ANOTHER "END CORRECTION" TO THIS SIMPLE MODEL.
(THE MOUTHPIECE CONE LENGTH $\ll$ THE WAVELENGTH BY DESIGN)

DEMO OPEN-CLOSED.WAV (LIKE CLARINET)
OPEN-OPEN.WAV (LIKE OBOE)

- YOU WILL ALSO SEE PIPE ORGANS USING ALL DIFFERENT SHAPES TO ADJUST TIMBRE
BASED ON THIS EFFECT.

VOICE FOR THIS CLASS, I WOULD VIEW THE VOICE AS A ^{HEAVILY} MODIFIED BRASS INSTRUMENT

- LUNGS PUSH AIR THROUGH VOCAL FOLDS (LIKE BUZZING LIPS)

- THROAT, MOUTH, AND SINUS CAVITIES ACT LIKE HIGHLY TUNABLE AND LOW-Q
RESONANT CHAMBERS (MORE LIKE HELMHOLTZ THAN PIPES)

THAT CAN ENHANCE SOME OSCILLATIONS, SUPPRESS OTHERS
+ AFFECT TIMBRE

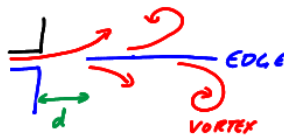
- IT'S SUPER INTERESTING + WOULD TAKE A CLASS TO PROPERLY EXPLORE!

- THESE TOOLS SHOULD REALLY HELP!

EDGE TONES

WIND IN TREES TO FLUTES & WHISTLES

- WHEN A JET OF AIR BLOWS AT AN EDGE, IT DOESN'T JUST SEPARATE! IT MAKES VORTICES & FLIPS UP & DOWN AT A CHARACTERISTIC FREQUENCY



$$f_{\text{EDGE}} \sim \frac{v_{\text{JET}}}{d}$$

← SPEED OF AIR LEAVING NOZZLE
← EDGE TONE FREQUENCY
← DISTANCE TO EDGE
FROM "FLUID DYNAMICS" PHYSICS COURSE (OOF!)

- THIS FLIPPING UP AND DOWN PRODUCES PRESSURE OSCILLATIONS → SOUND WAVES.
 - CAN HEAR THESE WHISTLES ON WINDY DAYS (WIND PICKS UP, PITCH INCREASES)
(OR RELATED)
(CAN HEAR THEM BLOWING ON THE EDGE OF A KNIFE IF YOU'RE CAREFUL)
- SO WHAT HAPPENS IF THIS FLIPPING RECEIVES FEEDBACK FROM A RESONANT CHAMBER? ☺

WHISTLE:



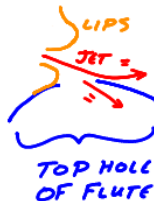
- THE AIR JET HERE BEHAVES A BIT LIKE A REED OR LIPS, BUT DOESN'T SO FORCEFULLY "SLAP" WHEN IT OSCILLATES, GIVING A DIFFERENT (MELLOWER?) TIMBRE. (PLUS THE PIPE-FILTERED "HISS" FROM THE WIND)
ULTRA BONUS: SIMULATE THIS ☺
- LIKE A REED, THE NATURAL EDGE TONE CAN BE MUCH HIGHER THAN THE BODY RESONANCE.
 - BUT WOW DOES THE STANDING WAVE PUSH THE JET! NOT SO STIFF AS A REED!

UNDERWHELMING DEMO: BLOW ON KNIFE EDGE & HEAR HIGH-PITCH WHISTLE.

2pt

Q Blowing _____ SELECTS HIGHER-FREQUENCY HARMONICS

FLUTE:



SAME AS FLUTE BUT LIPS PROVIDE JET RATHER THAN THE STRUCTURE ITSELF.

20pt

Q CAN YOU MAKE A FLUTE WITH BOTH ENDS OPEN?

ULTRABONUS 2: MAKE A CONICAL FLUTE. ☺
OR WHISTLE!